

Exploring the Unknown: A Survey on Embodied AI robotics in Uncharted Cluttered Environments

1st Houpu Gao

China Academy of Launch Vehicle Technology
Beijing, China
hg2624@columbia.edu

2nd Jialin Wang*

China Academy of Launch Vehicle Technology
Beijing, China
wangjl12@tsinghua.org.cn

3rd Benchang Zheng

China Academy of Launch Vehicle Technology
Beijing, China
sandstorm121@163.com

4th Jialin Zhu

China Academy of Launch Vehicle Technology
Beijing, China
zhu-jl19@tsinghua.org.cn

5th Hu Huang

Nanjing University
Nanjing, China
huanghu7@126.com

Abstract—Embodied Artificial Intelligence (EAI) robotics develop rapidly in recent years with the facilitation of Large Language Models (LLMs). The EAI robotics demonstrate high-autonomy capabilities in Uncharted Cluttered Environments (UCEs). In this paper, we provide a comprehensive overview of the system configuration and function modules of EAI robotics in UCEs. And the "Cerebrum-Cerebellum" framework is well-defined. Under this framework, the modules of perception, task planning and execution are analyzed roundly. Furthermore, we also present an overview of existing EAI robotics. Finally, we discuss the future directions of EAI robotics in UCEs. And our own thoughts and suggestions of technological approaches, capability boundaries and application scenarios are also given in this paper.

Index Terms—Embodied Artificial Intelligence, Embodied Cerebrum, Embodied Cerebellum, Robotics, Uncharted Cluttered Environments

I. INTRODUCTION

The rapid advancements in Artificial Intelligence (AI) and robotics have catalyzed the emergence of EAI, a paradigm that redefines the design and application of intelligent agents. Unlike traditional AI systems that rely solely on computational reasoning, EAI integrates physical interaction with the environment to achieve perception, decision-making, and task execution capabilities [1]. Early AI research prioritized symbolic reasoning and knowledge representation, whereas EAI emphasizes the interplay between computation and environmental interactions, advancing the autonomy and adaptability of robots [2].

Since the 1980s, research in EAI has progressed from theoretical foundations to practical applications [3]. Advances in sensing technologies, deep learning, and reinforcement learning have enabled robots to transition from executing predefined tasks to autonomous systems capable of perceiving, adapting, and optimizing their actions in dynamic and uncertain environments. Beyond cataloging algorithmic progress, it is now critical to assess their relative maturity, efficiency, and applicability under embodied constraints such as partial

observability, occlusions, resource limitations, and the sim-to-real gap.

Despite these advancements, there exist significant challenges in enabling robots to operate effectively in complex, dynamic, and uncertain environments [4]. For example, robots are required for advanced capabilities of real-time environmental perception, multi-source information processing, adaptive decision-making, and task execution under uncertainty in scenarios such as unexplored urban ruins, disaster zones, or extreme terrains.

The EAI agents are considered to possess these capabilities. To conceptualize the functional organization of EAI agents, we propose a "Cerebrum-Cerebellum" framework 1. In this framework, the "Cerebrum" encompasses high-level capabilities such as environment modeling, causal reasoning, and long-term task planning. The "Cerebellum" focuses on low-level functions, including sub-task execution, path planning, and real-time motion control, highlighting fast adaptation, robustness to perturbations, and graceful degradation under failures. Compared to traditional hierarchical control architectures, this metaphor emphasizes the dynamic interaction and continuous learning between high-level cognition and low-level execution, which is crucial for operating in complex, uncertain environments (UCEs).

The main contributions of this survey are summarized as follows:

- In this survey, we place emphasis on the high-autonomy capabilities of EAI robotics in uncharted cluttered dynamic environments. We provide a comprehensive overview of the systems configuration and function modules of agents with robotics in UCEs.
- We analyze perception, decision-making, and execution within the "Cerebrum-Cerebellum" framework, synthesizing their strengths and limitations, and outlining how their interaction supports resilient autonomy in UCEs.

*Corresponding author: Jialin Wang (wangjl12@tsinghua.org.cn)

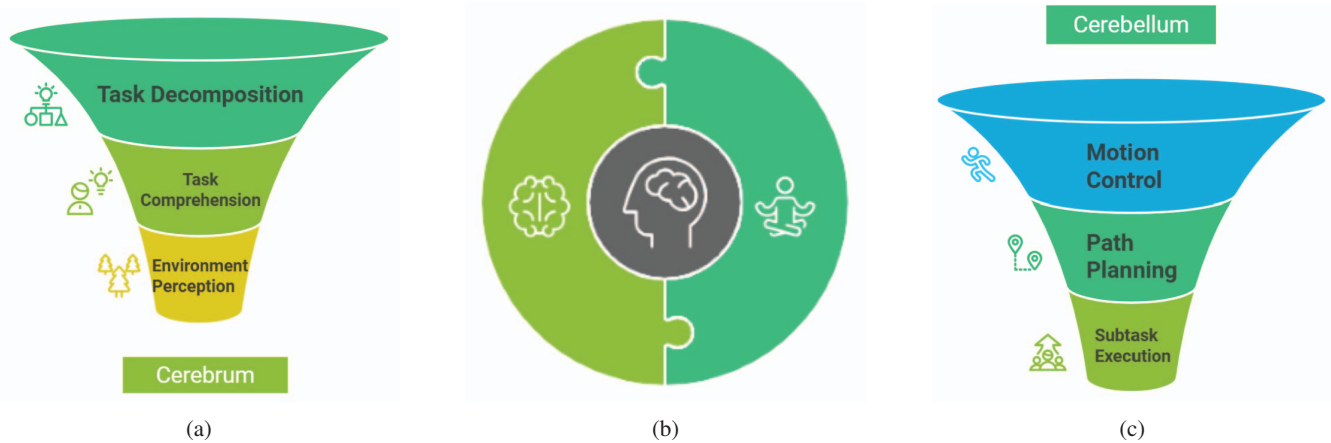


Fig. 1: Cerebrum-Cerebellum Framework

II. EMBODIED AI CEREBRUM: PLANNING

Planning in realistic physics environments can be regarded as the cerebrum in Embodied AI. The planning of complicated task in realistic physics environments, which is uncharted and cluttered, is confronted with great challenge. Thus, embodied perception, comprehension and planning are essential important compositions of embodied AI cerebrum.

A. Embodied Environment Perception: Object Detection

Object detection is a necessary processing to recognize the world, which can be divided into close world object detection and open world object detection. The close world object detection is based on a predefined, fixed set of object categories. The open world object detection methods are expected to have the ability to detect and cope with unknown objects, which is much more natural in the real world.

1) *Close World Object Detection*: The traditional deep learning based object detection is data driven. The categories of these methods can be divided as below: 1. Two-stage methods; 2. One-stage methods; 3. The transformer based methods.

The two-stage object detection methods first perform region proposals (RP, a pre-selected box that may contain the objects to be detected). Then samples classification is implemented by a convolutional neural network (CNN). There are some common two-stage object detection algorithms, such as R-CNN [5], SPP-Net [6], Faster R-CNN [7], etc. Two-stage based methods have high accuracy in object detection. However, the computational and time costs of these methods are very huge, on account of the RP generation and processing. As a result, two-stage based object detection methods are not suitable for scenarios with high real-time requirements.

One-stage object detection methods implement the object detection task as a single regression problem, without the generation of RPs. The computational and time costs of one-stage based methods are lower than two-stage based methods. However, the accuracy of these methods will decrease in the complex scene, small targets or existence of occlusion among

the targets, etc. Therefore, the one-stage object detection methods, such as YOLOv3 [8], other members of YOLO family [9], SSD (Single Shot Detector) [10], etc, can satisfy the application scenarios with high real-time requirements.

Transformer demonstrated powerful ability of data processing in natural language processing. Transformer based object detection methods can capture global features of images by introducing transformer into object detection [11]. The self-attention mechanism is beneficial for Transformer to understand the interrelationship between complex scenes and objects. Moreover, the architecture of Transformer is flexible and can be combined and extended with other network structures easily. Therefore, the end-to-end object detection with Transformers (DETR) was proposed [12]. There are some improved versions of DETR method, such as Deformable DETR [13], Dn-DETR [14], etc. However, due to the high computational complexity of the self-attention mechanism, Transformer-based methods require a large amount of computational resources. These methods are more suitable for object detection in complex scenes or with multi-modal data fusion.

2) *Open World Object Detection*: The models in open world object detection methods are required to have generalization ability, which can process objects with unknown categories reasonably. Therefore, the open world object detection methods are more suitable for realistic scenarios due to the variety of categories in the real world. The common point of these methods is to add semantic information during the process of model training to enable themselves detect unknown categories.

Li et.al. proposed [15] the grounded language-image pre-training (GLIP) method, which first applied CLIP model in object detection. GLIP integrated semantic information into the process of model training, in which the text encoder is the BERT model [16] and the visual encoder is the swin model [17]. In addition, the text and visual features have deep interactions in GLIP, by adding extra BERT layers and DyHead layers behind the text and visual encoder respectively. Then the dictionary-enriched visual-concept paralleled pre-

training (DetCLIP) method was proposed for open world object detection [18], which improved the input of text information.

In recent years, more open world object detection methods were proposed. Gu et.al. proposed the Vision and Language Knowledge Distillation (ViLD) method, which adopts distillation learning to construct the model and utilizes the architecture of both encoder and decoder models of CLIP [19]. The OWLv2 [20] was also proposed on the basis of Transformer. Then the Grounding DINO was proposed [21], in which the text information is assumed on the basis of DINO. With the increasing popularity of the multimodal models, a variety of open world object detection methods have been proposed, such as the Open World DETR (OW-DETR) [22], Detecting Everything in the Open World (UniDetector) [23], YOLO-World [24], etc. YOLO-World shows high real-time performance inheriting from the lightweighting characteristics of the YOLO family. Owl-vit performs strong generalization on open vocabulary and long-tailed distribution datasets. GroundingDINO excels in semantic understanding of images and text, which is able to detect objects quickly and accurately.

B. Embodied Comprehension and Planning

1) *Multi-modal Large Models:* Multimodal Large Models (MMLMs) are one of the top research aspects in Artificial Intelligence (AI), which enables machine to understand and fuse image, point cloud, text and other types of information. Recently, MMLMs show great application potential in many fields, such as intelligent image description generation, visual question and answer system, image semantic understanding and retrieval, autonomous navigation, etc. According to the fusion approaches of multi modal information, MMLMs can be categorized into early fusion at the input stage and modality fusion within the internal layers [25].

The MMLMs of early fusion at the input stage allow models to be exposed to multimodal information at an initial stage, helping to learn a joint representation across modalities, such as CLIP [26], DALL-E 2 [27], LEO [28], etc. Huang et.al. proposed LEO, in which multimodal data is converted into token sequences and then processed by a variety of embedding models, fed into the LLM for further processing. DALL-E 2 is constructed for image generation, in which the results of text encoder and image encoder are combined for the processing flow of the model in the architecture of DALL-E 2. The early fusion MMLMs first merge data from different modalities directly in the subsequent processing of the model. This approach is able to fully utilize the original correlations between different modal data by mining the richest inter-modal interaction information. However, this approach requires high quality of the preprocessing of different modal data. Moreover, it is obsessive sensitive of a modal data noise or poor quality.

There contain fusion modules of specific architecture in MMLMs of modality fusion within the internal layers. In MMLMs of this fusion approach, data of different modalities are input into different branches or modules to be processed individually, such as feature extraction and transformation.

Then the processing results of multi-modal data are fused through a particular fusion mechanism (e.g., based on the attentional mechanism, weighted summation, etc.). The typical MMLMs of modality fusion within the internal layers include UNITER (Universal Image-Text Representation Learning) [29], SkipCrossNets [30], MmAP (Multi-modal Alignment Prompt) [31], etc. UNITER [29] was proposed for learning generic image-text representation. In UNITER, the visual and language information integrate deeply by a Transformer-based architecture with its multi-head attention mechanism. It achieves good performance on several visual language tasks. This fusion approach can deeply integrate features from different modalities. Therefore, this type of MMLMs performs better in tasks which require in-depth mining of semantic relationships among different modalities. However, the fusion modules can enhance the difficulty of model designing and training.

2) *Task Comprehension and Planning:* Embodied task comprehension and planning can be divided into two main parts: semantic comprehension and intent resolution, target decomposition and subtasks generation.

In semantic comprehension and intent resolution, analyzing the actual content and desired conveyed goals is the key issue. It relies on lexical semantics and grammatical structure analysis to clarify the meanings and interrelationships of different parts. It is also necessary to mine out the implied conditions, desired outcomes and categories of tasks. Then the embodied agents can plan their actions according to the semantic comprehension and intent resolution results. Knowledge graphs (KGs) play an important role in embodied semantic understanding and intent resolution in result of that they contain a large number of entities and relationships of these entities. Then KGs can assist the embodied agent in making sensible decisions in this approach. Qi et al. [32] proposed the safety control of service robots with embodied KGs, in which the KG can help ensure safety while correctly interpreting task instructions.

Goal decomposition and subtasks generation also play an important role in task comprehension and planning, which can decompose a complex task into multiple executable subtasks. The goal decomposition and subtasks generation methods are mainly based on intelligent optimization algorithms or Large models (LMs). Intelligent optimization algorithms include Partially Observable Markov Decision Process (POMDP) planning [33] and heuristic search algorithms (e.g. genetic algorithm [34]). Due to the partial observability of the environment, POMDP can be utilized to generate the optimal sequence of subtasks according to the current observation and historical information. The heuristic search algorithms can be applied in task space searching. Then the feasible subtasks sequences can be generated. Relying on the powerful language understanding and generation capabilities of LMs, the comprehension of task instructions and environmental information can be improved, such as the DECKARD agent [35] and VoxPoser [36]. In both DECKARD and VoxPoser, LLMs decompose complex operational tasks into explicit subtasks

and determine the order of subtasks to be performed. Thus the complex problem can be refined into actionable steps, allowing the robot to complete the task in an organized manner. LMs perform satisfactorily in goal decomposition and subtasks generation, due to their capabilities of powerful semantic understanding, efficient reasoning and planning, zero-shot or few-shot learning ability.

III. EMBODIED AI CEREBELLUM: EXECUTION

In uncharted and cluttered environments, the execution of sophisticated pre-planned actions requires precise coordination, real-time adaptability, and robustness to dynamic changes. To achieve this, motion control provides the fundamental actuation capabilities of the agent, while path planning ensures the execution of intricate action sequences.

A. Motion Control

Motion control refers to controlling a robot's driving components to ensure that it executes the actions planned by the cerebrum while maintaining smoothness, stability, and safety. This process involves using suitable control strategies to adjust actuators based on real-time sensor feedback. We focus on recent learning-based motion control methods in this subsection.

1) *Imitation Learning-based Method*: Imitation learning enables robots to acquire control policies by imitating expert behaviors. The model is trained to replicate continuous human-like actions using demonstration data. This approach is suited for tasks that are difficult to model with traditional control methods. Current imitation learning methods can be broadly categorized into Behavioral Cloning (BC) and Inverse Reinforcement Learning (IRL).

Behavioral Cloning (BC) [37] maps expert demonstration data into state-action pairs and uses supervised learning to train the model. While BC is easy to implement, it has notable limitations, including its reliance on finite expert demonstrations, which restrict its ability to generalize to out-of-distribution (OOD) scenarios. To address these challenges, SafeDagger [38] introduced safety constraints during training to minimize risks and ensure robust performance in OOD scenarios.

Inverse Reinforcement Learning (IRL) infers the underlying reward function from observed expert behaviour and uses it to derive optimal policies. This method is suited for tasks with complex or multi-modal objectives where explicit reward design is infeasible. Generative Adversarial Imitation Learning (GAIL) [39] is a prominent algorithm in this domain, which leverages a discriminator network as a reward function to guide policy learning. GAIL enables the derivation of optimal policies aligned with inferred goals and reward structures.

2) *Deep Reinforcement Learning-based Method*: Deep Reinforcement Learning (DRL) integrates Deep Learning and Reinforcement Learning, enabling robots to learn and optimize policies in uncharted, cluttered environments through interaction. Unlike traditional methods, DRL does not rely on explicit models, making it effective for dynamic adjustment and action

optimization. DRL can be broadly categorized into online DRL and offline DRL.

On-policy DRL refers to a learning method in which the agent updates its policy based on the data collected by following its current policy. On-policy DRL ensures stable updates but suffers from low sample efficiency, which limits its application in complex environments. Policy Gradient (PG) [40] is a representative method that optimizes policy performance through On-policy DRL. PG is effective in continuous, high-dimensional action spaces but has low sample efficiency, slow convergence, and high sensitivity to reward signals. Actor-Critic methods address these limitations by separating decision-making (Actor) from value evaluation (Critic). A key algorithm is Asynchronous Advantage Actor-Critic (A3C) [41], which uses parallel agents to collect data and asynchronously update the policy. While A3C significantly improves efficiency, its stability is still needed when dealing with complex tasks. Proximal Policy Optimization (PPO) [42] introduced a constraint in the objective function that limits the difference between the new and old policies.

Off-policy DRL allows the agent to learn from data generated by a different policy, which can be more exemplary. Off-policy approaches can learn from the historical data generated by a behavior policy that differs from the target policy, enabling efficient reuse of experience. Off-policy DRL improves sample efficiency, making it ideal for environments where real-time interaction is costly or infeasible, such as Q-Learning [43], (DDPG) [44], Soft Actor-Critic (SAC) [45], etc.

Offline DRL learns policies from a pre-collected dataset without interacting with the environment during training. This approach is ideal for scenarios where real-time interaction is costly, risky, or impractical. Recent advancements have introduced Diffusion Policy [46] as a promising method within offline DRL for robot motion control, which is well-suited for handling high-dimensional continuous action spaces and dynamic environments. 3D Diffuser Actor [47] integrated diffusion policies with 3D scene feature representations and language-based instructions to optimize. It can demonstrate robust zero-shot generalization to unseen tasks and environments.

B. Path Planning

Path planning aims to determine a collision-free path from a starting position to a goal position in uncharted, cluttered environments, balancing optimality, safety, and feasibility.

1) *Globally*: Search-based methods explored feasible paths in the configuration space by constructing graphs or trees. These methods alternate between graph construction and search, selecting the best node for expansion at each iteration based on a predefined strategy [48]. Search-based path planning methods are suitable for environments that discrete representations can effectively represent. A* Algorithm [49] integrated Dijkstra's algorithm with heuristic evaluation and calculated the total cost by combining the actual cost from the start node to the current node with an estimated cost from the current node to the goal. Despite its versatility

and accuracy, A* has high computational demands. Various variants of A* have been developed to enhance efficiency: Bidirectional A* Algorithm [50] searched simultaneously from the start and goal; Real-time Adaptive A* (RTAA*) [51] updated the heuristic function during search dynamically.

Machine Learning (ML) enhances traditional path planning methods' computational efficiency and flexibility by automatically learning effective planning strategies, particularly in high-dimensional spaces. Furthermore, [52] employed deep learning models to predict the volume swept by a robot, significantly reducing computational costs for collision detection and accelerating planning. ML can also identify optimal combinations of methods for specific decision-making tasks or propose alternative solutions when existing paths prove unfeasible [53].

2) *Locally*: Optimization methods refine an initial path locally to enhance smoothness, avoid obstacles, and reduce energy or time consumption.

Covariant Hamiltonian Optimization for Motion Planning (CHOMP) [54] is a gradient-based local optimization method. CHOMP minimized the trajectory's cost function by leveraging the Covariant Hamiltonian concept. K-Order Markov Optimization (KOMO) [55] formulated the planning problem as a multi-stage optimization process. KOMO is effective in complex scenarios but incurs high computational costs, particularly in high-dimensional spaces. Graph of Convex Sets (GCS) [56] represented the path planning problem as a graph of convex sets, optimizing paths by searching within this graph. GCS guarantees optimal solutions within convex regions, providing efficient obstacle avoidance.

IV. EMBODIED AI ROBOTS FOR UNKNOWN CLUTTERED ENVIRONMENTS

In this section, we introduce several typical embodied AI robots, including DARPA Subterranean (SubT) Challenge and its champion team CERBERUS, MIT Cheetah and NASA Mars Rovers, which are summarized in table I.

V. CONCLUSIONS

This survey systematically provides a comprehensive statement of Embodied AI robotics in uncharted cluttered environments through the lens of the "Cerebrum-Cerebellum" framework. By decoupling high-level cognition (perception, reasoning, planning) from low-level execution (motion control, real-time adaptation), we establish a unified paradigm for analyzing autonomous systems.

Future research should prioritize:

- Neuromorphic Sensor-Processor Co-design for energy-efficient multimodal perception.
- Causal Reasoning Modules to improve task decomposition in partially observable states.
- Federated Embodied Learning frameworks enabling collective intelligence across robot swarms.
- Ethical AI Governance protocols for responsible autonomy in safety-critical UCE applications.

In the future, we will focus on the research of the path toward truly adaptive EAI systems of robotics, enabling their rapid applications in uncharted cluttered environments.

REFERENCES

- [1] J. Duan, S. Yu, H. L. Tan, H. Zhu, and C. Tan, "A survey of embodied ai: From simulators to research tasks," 2022.
- [2] Y. Hu, Q. Xie, V. Jain, J. Francis, J. Patrikar, N. Keetha, S. Kim, Y. Xie, T. Zhang, H.-S. Fang, S. Zhao, S. Omidshafiei, D.-K. Kim, A. Akbar Agha-mohammadi, K. Sycara, M. Johnson-Roberson, D. Batra, X. Wang, S. Scherer, C. Wang, Z. Kira, F. Xia, and Y. Bisk, "Toward general-purpose robots via foundation models: A survey and meta-analysis," 2024.
- [3] R. Pfeifer and F. Iida, "Embodied artificial intelligence: Trends and challenges," in *Embodied Artificial Intelligence*, 2003.
- [4] Y. Ali, "Autonomous systems: Challenges and opportunities," *Advances in Engineering Innovation*, vol. 4, pp. 33–37, 11 2023.
- [5] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 580–587, 2014.
- [6] P. Purkait, C. Zhao, and C. Zach, "Spp-net: Deep absolute pose regression with synthetic views," *arXiv preprint arXiv:1712.03452*, 2017.
- [7] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *IEEE transactions on pattern analysis and machine intelligence*, vol. 39, no. 6, pp. 1137–1149, 2016.
- [8] A. Vidyavani, K. Dheeraj, M. R. M. Reddy, and K. N. Kumar, "Object detection method based on yolov3 using deep learning networks," *Int. J. Innov. Technol. Explor. Eng.*, vol. 9, no. 1, pp. 1414–1417, 2019.
- [9] M. A. R. Alif and M. Hussain, "Yolov1 to yolov10: A comprehensive review of yolo variants and their application in the agricultural domain," *arXiv preprint arXiv:2406.10139*, 2024.
- [10] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "Ssd: Single shot multibox detector," in *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*, pp. 21–37, Springer, 2016.
- [11] J. Beal, E. Kim, E. Tzeng, D. H. Park, A. Zhai, and D. Kislyuk, "Toward transformer-based object detection," *arXiv preprint arXiv:2012.09958*, 2020.
- [12] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *European conference on computer vision*, pp. 213–229, Springer, 2020.
- [13] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable detr: Deformable transformers for end-to-end object detection," *arXiv preprint arXiv:2010.04159*, 2020.
- [14] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "Dn-detr: Accelerate detr training by introducing query denoising," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 13619–13627, 2022.
- [15] L. H. Li, P. Zhang, H. Zhang, J. Yang, C. Li, Y. Zhong, L. Wang, L. Yuan, L. Zhang, J.-N. Hwang, et al., "Grounded language-image pre-training," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10965–10975, 2022.
- [16] J. Devlin, "Bert: Pre-training of deep bidirectional transformers for language understanding," *arXiv preprint arXiv:1810.04805*, 2018.
- [17] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 10012–10022, 2021.
- [18] L. Yao, J. Han, Y. Wen, X. Liang, D. Xu, W. Zhang, Z. Li, C. Xu, and H. Xu, "Detclip: Dictionary-enriched visual-concept paralleled pre-training for open-world detection," *Advances in Neural Information Processing Systems*, vol. 35, pp. 9125–9138, 2022.
- [19] X. Gu, T.-Y. Lin, W. Kuo, and Y. Cui, "Open-vocabulary object detection via vision and language knowledge distillation," *arXiv preprint arXiv:2104.13921*, 2021.
- [20] M. Minderer, A. Gritsenko, and N. Houlsby, "Scaling open-vocabulary object detection," *Advances in Neural Information Processing Systems*, vol. 36, 2024.

TABLE I: Comparison of Embodied AI Systems for Exploration.

Robot Name	Strengths	Limitations	Key Capabilities
CERBERUS (SubT)	- Multi-robot collaboration - Robust in extreme environments - Multi-modal perception - RL-based control	- Reliance on wireless comm. - Limited maneuverability in tight spaces - Complex system integration	- SLAM - Autonomous navigation - Search and rescue - Environmental mapping - Multi-robot coordination
MIT Cheetah	- Lightweight design - High-speed mobility - RL for motion optimization	- Limited payload capacity - Poor adaptability in cluttered terrain	- Agile locomotion - Real-time path planning - Obstacle avoidance
NASA Mars Rovers	- Long-duration autonomy - Radiation resistance - Precise terrain mapping	- Slow traversal speed - High communication latency	- Sensor fusion 3D environmental modeling - Hierarchical planning - Adaptive terrain traversal

- [21] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su, *et al.*, “Grounding dino: Marrying dino with grounded pre-training for open-set object detection,” in *European Conference on Computer Vision*, pp. 38–55, Springer, 2025.
- [22] A. Gupta, S. Narayan, K. Joseph, S. Khan, F. S. Khan, and M. Shah, “Ow-detr: Open-world detection transformer,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 9235–9244, 2022.
- [23] Z. Wang, Y. Li, X. Chen, S.-N. Lim, A. Torralba, H. Zhao, and S. Wang, “Detecting everything in the open world: Towards universal object detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11433–11443, 2023.
- [24] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, “Yolo-world: Real-time open-vocabulary object detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16901–16911, 2024.
- [25] S. N. Wadekar, A. Chaurasia, A. Chadha, and E. Culurciello, “The evolution of multimodal model architectures,” *arXiv preprint arXiv:2405.17927*, 2024.
- [26] A. Radford and J. W. e. a. Kim, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*, pp. 8748–8763, PMLR, 2021.
- [27] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, “Hierarchical text-conditional image generation with clip latents,” *arXiv preprint arXiv:2204.06125*, vol. 1, no. 2, p. 3, 2022.
- [28] J. Huang, S. Yong, X. Ma, X. Linghu, P. Li, Y. Wang, Q. Li, S.-C. Zhu, B. Jia, and S. Huang, “An embodied generalist agent in 3d world,” *arXiv preprint arXiv:2311.12871*, 2023.
- [29] Y.-C. Chen, L. Li, L. Yu, A. El Kholy, F. Ahmed, Z. Gan, Y. Cheng, and J. Liu, “Uniter: Universal image-text representation learning,” in *European conference on computer vision*, pp. 104–120, Springer, 2020.
- [30] Y. Gong, X. Zhang, H. Liu, X. Jiang, Z. Li, X. Gao, L. Lin, D. Jin, J. Li, and H. Liu, “Skipcrossnets: Adaptive skip-cross fusion for road detection,” *arXiv preprint arXiv:2308.12863*, 2023.
- [31] Y. Xin, J. Du, Q. Wang, K. Yan, and S. Ding, “Mmap: Multi-modal alignment prompt for cross-domain multi-task learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 16076–16084, 2024.
- [32] Y. Qi, G. Kyebambo, S. Xie, W. Shen, S. Wang, B. Xie, B. He, Z. Wang, and S. Jiang, “Safety control of service robots with llms and embodied knowledge graphs,” *arXiv preprint arXiv:2405.17846*, 2024.
- [33] M. Lauri, D. Hsu, and J. Pajarinen, “Partially observable markov decision processes in robotics: A survey,” *IEEE Transactions on Robotics*, vol. 39, no. 1, pp. 21–40, 2022.
- [34] Y. Xu, K. Li, J. Hu, and K. Li, “A genetic algorithm for task scheduling on heterogeneous computing systems using multiple priority queues,” *Information Sciences*, vol. 270, pp. 255–287, 2014.
- [35] K. Nottingham, P. Ammanabrolu, A. Suhr, Y. Choi, H. Hajjishirzi, S. Singh, and R. Fox, “Do embodied agents dream of pixelated sheep: Embodied decision making using language guided world modelling,” in *International Conference on Machine Learning*, pp. 26311–26325, PMLR, 2023.
- [36] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, “Voxposer: Composable 3d value maps for robotic manipulation with language models,” *arXiv preprint arXiv:2307.05973*, 2023.
- [37] A. Mandlekar, D. Xu, J. Wong, S. Nasiriany, C. Wang, R. Kulkarni, L. Fei-Fei, S. Savarese, Y. Zhu, and R. Martín-Martín, “What matters in learning from offline human demonstrations for robot manipulation,” 2021.
- [38] J. Zhang and K. Cho, “Query-efficient imitation learning for end-to-end autonomous driving,” 2016.
- [39] J. Ho and S. Ermon, “Generative adversarial imitation learning,” 2016.
- [40] M. Lehmann, “The definitive guide to policy gradients in deep reinforcement learning: Theory, algorithms and implementations,” 2024.
- [41] V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. P. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu, “Asynchronous methods for deep reinforcement learning,” 2016.
- [42] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” 2017.
- [43] C. Watkins and P. Dayan, “Q-learning,” *Machine Learning*, vol. 8, pp. 279–292, 1992.
- [44] D. Silver, G. Lever, N. M. O. Heess, T. Degris, D. Wierstra, and M. A. Riedmiller, “Deterministic policy gradient algorithms,” in *International Conference on Machine Learning*, 2014.
- [45] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” 2018.
- [46] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” 2024.
- [47] T.-W. Ke, N. Gkanatsios, and K. Fragkiadaki, “3d diffuser actor: Policy diffusion with 3d scene representations,” 2024.
- [48] G. Sotirchos and Z. Ajanovic, “Search-based versus sampling-based robot motion planning: A comparative study,” 2024.
- [49] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Transactions on Systems Science Cybernetics*, vol. 4, no. 2, pp. 28–29, 1972.
- [50] S. Sharma, S. Srijan, and J. V. Vidhya, *Parallelizing Bidirectional A* Algorithm*. Intelligent Systems and Computer Technology, 2020.
- [51] C. Hernández and J. Baier, “Real-time adaptive a* with depression avoidance,” 01 2011.
- [52] H.-T. L. Chiang, A. Faust, S. Sugaya, and L. Tapia, “Fast swept volume estimation with deep learning,” in *Workshop on the Algorithmic Foundations of Robotics*, 2018.
- [53] S. Li and N. T. Dantam, “Learning proofs of motion planning infeasibility,” *Robotics: Science and Systems XVII*, 2021.
- [54] M. Zucker, N. R. Google, A. D. Dragan, M. Pivtoraiko, M. Klingensmith, N. D. Ratliff, C. M. Dellin, J. A. Bagnell, S. S. Srinivasa, A. D. Dragan, S. S. Srinivasa, P. Pa, and P. Pa, “Chomp: Covariant hamiltonian optimization for motion planning,” *The International Journal of Robotics Research*, vol. 32, pp. 1164 – 1193, 2013.
- [55] M. Toussaint, “Newton methods for k-order markov constrained motion problems,” 2014.
- [56] T. Marcucci, J. Umenberger, P. Parrilo, and R. Tedrake, “Shortest paths in graphs of convex sets,” *SIAM Journal on Optimization*, vol. 34, p. 507–532, Feb. 2024.